

Model Card — Hate Speech Detection Classifier

TF-IDF + Logistic Regression for binary hate-speech detection on the ETHOS dataset

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1. Model details

Field	Description
Model name	ETHOS-Hate-v1
Task	Binary classification of English online comments: Hate / Non-Hate.
Model	TF-IDF word unigrams + bigrams, followed by Logistic Regression (class_weight=balanced, max_iter=2000).
Dataset	ETHOS binary dataset; 998 comments (433 hate, 565 non-hate). Kaggle dataset: konradb/ethos-hate-speech-dataset.
Train / test	Stratified 80/20 split: 798 training comments and 200 held-out test comments; random_state=42.
Preprocessing	Missing/empty text and exact duplicates removed; lowercase and Unicode accent normalization handled by TF-IDF.
Tools	Python / Google Colab, KaggleHub, Pandas, NumPy, scikit-learn, Matplotlib and Seaborn.

2. Intended use

- Intended:** Educational first-pass flagging of potentially hateful English comments for review by a human moderator.
- Out of scope:** Automatic content removal, bans or reports without human review; non-English or code-mixed text; production decisions based on this model alone.

3. Overall performance

Metric	Result
Accuracy	0.695 (69.5%)
Hate precision / recall	0.676 / 0.575
Hate F1 / Macro F1	0.621 / 0.683
ROC-AUC	0.727
Majority-class baseline	56.5% accuracy (always predicting Non-Hate)

On the 200-comment test set, 139 predictions were correct. The model detected 50 of 87 hate comments and missed 37, so accuracy alone does not capture the moderation risk.

4. Performance across text subgroups

The audit compares short vs. long comments and comments containing profanity vs. those without it. These are text-style groups, not a demographic fairness audit.

Subgroup	n	Accuracy	Hate precision	Hate recall	F1
Short (≤14 words)	104	68.3%	61.0%	59.5%	60.2%
Long (>14 words)	96	70.8%	75.8%	55.6%	64.1%
Profanity	31	67.7%	68.0%	89.5%	77.3%
No profanity	169	69.8%	67.3%	48.5%	56.4%

Key finding: Hate recall is 89.5% with profanity but only 48.5% without it. Of 12 non-hate comments containing profanity, 8 were falsely flagged (67%). This suggests a possible shortcut association, but the profanity subgroup is small (n=31).

5. Explainability check

Method: inspect Logistic Regression coefficients for TF-IDF features. Positive values push the score towards Hate; negative values push it towards Non-Hate.

Hate-leaning feature	Weight	Non-hate-leaning feature	Weight
ass	+1.4239	love	-1.1532
kill	+1.1557	can	-1.0982
gays	+1.1343	me	-1.0592
fuck	+1.1036	the	-1.0219
stupid	+1.0709	am	-0.9393

These coefficients show learned associations, not intent. Identity-related terms may appear in harmless contexts and can contribute to false flags.

6. Limitations and ethical considerations

- Small dataset (998 comments) and only 200 test examples; subgroup estimates may be noisy.
- Hate recall is 57.5%, meaning the model misses 37 of 87 hate comments.
- TF-IDF + Logistic Regression does not deeply model intent, context, sarcasm, or negation; results are English-only.
- Profanity and identity-related terms may act as shortcuts. No demographic fairness audit was performed.

7. Recommendations

- Use only as a first-pass moderation aid with human review; do not automate punitive action.
- Evaluate on a larger dataset, use cross-validation, and compare contextual models such as BERT/RoBERTa.
- Extend auditing to identity-term swaps and other writing styles; investigate false positives and false negatives.

References: ETHOS Hate Speech Dataset (Kaggle); Mollas et al., “ETHOS: an Online Hate Speech Detection Dataset” (2020); scikit-learn documentation for TfidfVectorizer and LogisticRegression. Figures from the project notebook (random_state=42).